

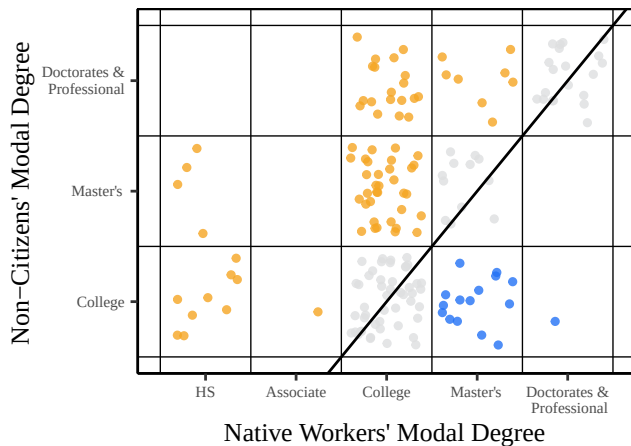
Discounted Degrees: Employer Frictions and the Misallocation of High-Skilled Immigrants

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Empirical Micro Lunch, Oct.1 2026

Immigrants hold more schooling than citizens in many of the same occupations



Each dot is an occupation.

- 75 occupations: immigrants have **more schooling**
- 80: same modal degree
- 16: immigrants have **less schooling**

The paper asks whether employer-side costs help produce these assignments.

ACS 2005–2024, excluding 2020. Non-citizens whose highest degree is from a US institution, identified by immigration year and education age.

Why do high-skilled immigrants take jobs that use less education than they have?

- The literature documents immigrant mismatch and wage penalties.
- The unresolved issue is **why the match changes**: information, credential transferability, licensing, or sponsorship costs.
- I use career histories and a work-authorization reform to separate descriptive placement patterns from a policy effect.

Wage constraints shift adjustment from pay to job assignment

- ① Hiring an immigrant can cost the firm more through sponsorship, delay, risk, or discrimination.
- ② If the firm cannot pass that cost through to pay for the same job, it changes the contract it offers. Visa rules set a wage floor: the prevailing wage at H-1B conversion, which employers anticipate at OPT hiring.
- ③ The model predicts two responses:
 - **Under-placement:** keep the degree, take a job requiring less education.
 - **Over-signaling:** acquire more schooling to reach a matching job.

Entry pattern: excess over-education for immigrant master's holders; PhDs much closer to parity.

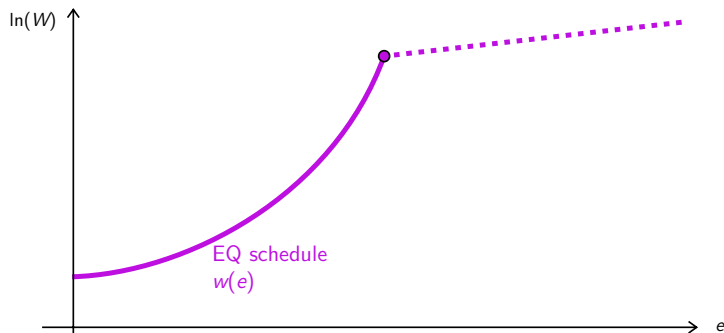
Main findings: Immigrants' jobs, not their pay, absorb the cost of hiring them.

- ① **At entry:** master's holders are 2.5 pp (US degree) and 6.9 pp (foreign degree) more over-educated than natives with the same degree, tier, and field. PhDs are near parity.
- ② **Over ten years:** no common catch-up. Foreign-degree master's holders stay behind; foreign-degree PhDs close most of their gap.
- ③ **Policy:** the 2008 STEM-OPT extension added about 3 US first jobs per 100 STEM graduates.

Contribution: a wage floor turns a hiring friction into job assignment

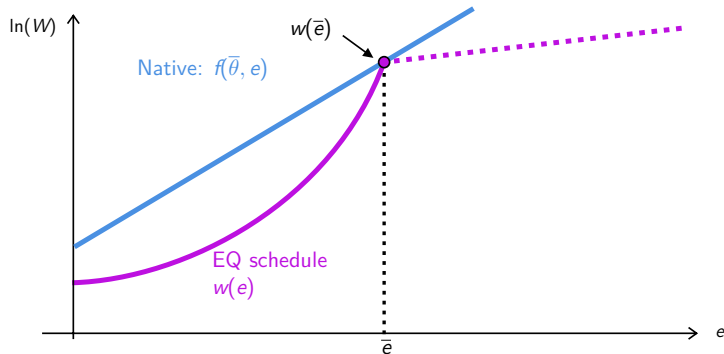
	What we know	What I add
Immigrant over-education	Persistent gap; employer screening proposed, not tested (Chiswick & Miller 2009; Cassidy & Gaulke 2024).	A model of screening and a test: master's under-placed, PhDs near parity.
Signaling with a group friction	Friction on Black workers acts on the <i>wage</i> ; they get more schooling (Lang & Manove 2011).	Wage floors block that, so the friction moves the <i>job</i> .
High-skill visa policy	Effects on firms, on who stays, and on cap-exempt careers (Doran, Gelber & Isen 2022; Demirci 2019; Amuedo-Dorantes & Furtado 2019).	Whether the jobs match their education .
Foreign credentials	Foreign schooling valued below domestic schooling (Friedberg 2000).	I separate workers by where the degree was earned.
Assimilation	Cross-sections mix cohorts and return migration (Borjas 1985; Lubotsky 2007).	Fixed graduation cohorts; balanced stayer sample.

Conceptual Framework



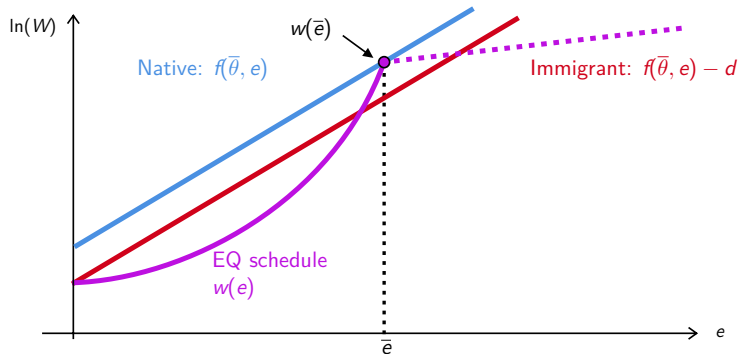
- $w(e')$: value of the job typically held with e' .

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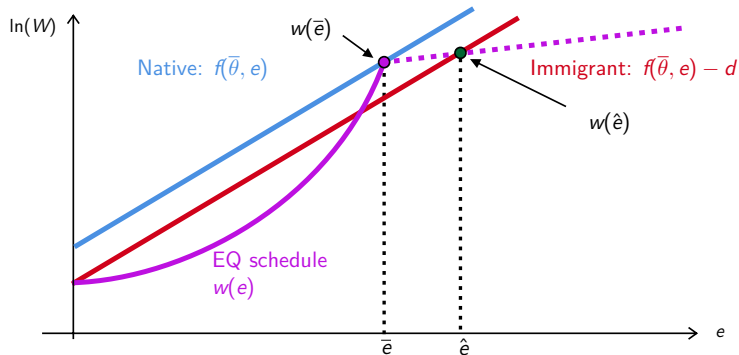
- $w(e')$: value of the job typically held with e' .
- Dotted schedule is rationalized by compensating wage differentials.
- $\bar{e}$: what the most able native chooses (think: master's).

Conceptual Framework



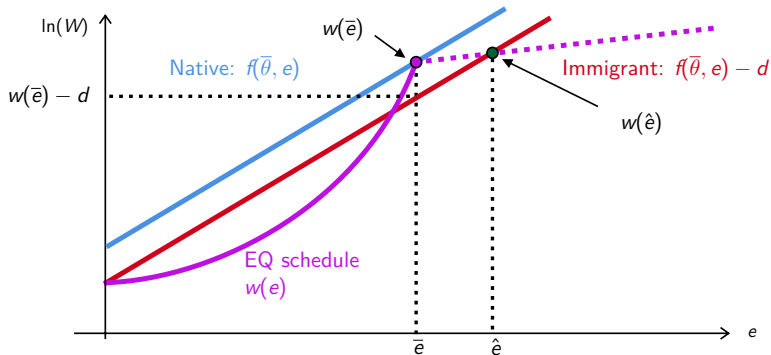
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- d : what the immigrant costs the firm beyond a native.

Conceptual Framework



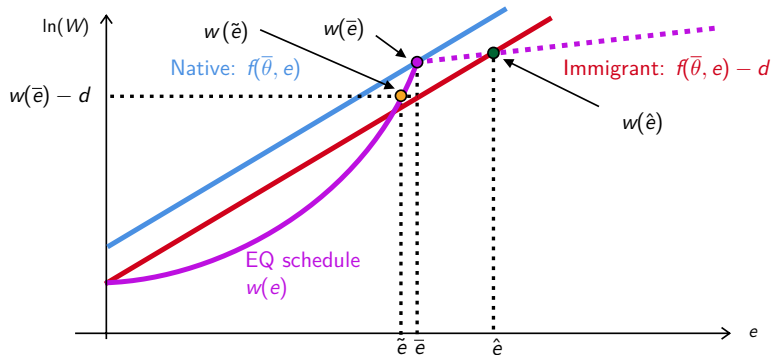
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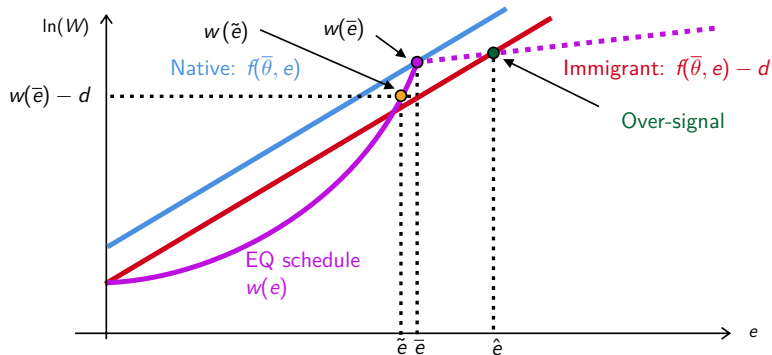
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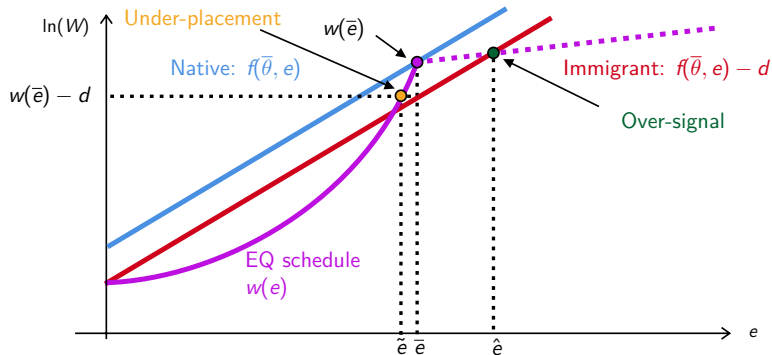
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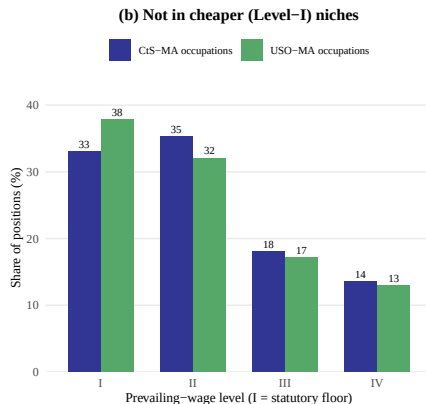
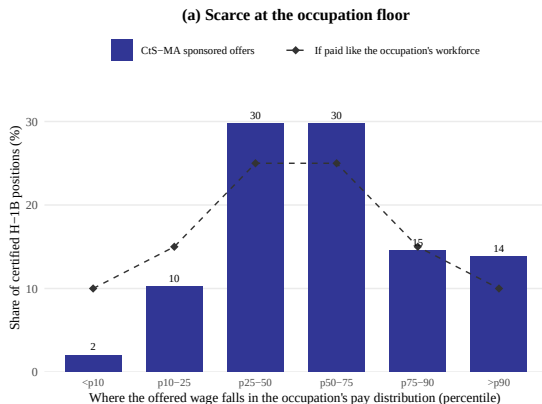
- ① Same ability, but the immigrant is worth d **less** to the firm, and the firm can't cut their pay.
- ② So they end up in one of two places:
 - **Under-placement:** keeps their master's, takes a job meant for less education $\Rightarrow$ *over-educated*.
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- ③ **Test:** immigrant master's holders should be over-educated more often than natives; immigrant doctorates should be at or near parity.

Visa wage floors bind: sponsored master's holders are rarely paid below the occupation's 25th percentile



Certified H-1B LCA filings, FY2011–2025, position-weighted; master's-degree entrants. The benchmark is the occupation's incumbent wage distribution, not matched native entrants.

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Testing the mechanism

- ③ **Policy response.** Lowering the hiring friction should increase US placement and improve the jobs held by workers who remain.

- LinkedIn profiles (People Data Labs): full education and job histories.

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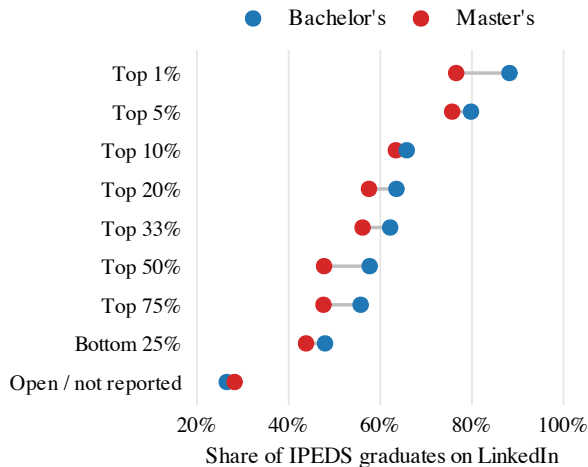
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Title	Predicted occupation
senior page designer and copy editor	Editors
international visiting attorney	Lawyers
"web engineer" (in Chinese)	Computer Systems Engineers/Architects
"PhD student" (in Chinese)	Teaching Assistants, Postsecondary
gfliwn askdm	rejected (00-0000)

LinkedIn over-samples selective schools and corporate fields



- **56%** of US bachelor's and **51%** of master's graduates are on LinkedIn.

- Bachelor's: **26%** at open-admission schools, **88%** at the top 1%.

- Business, engineering over; education, health under.

Selection biases my results only if who joins LinkedIn depends on job outcomes differently for the two groups.

Pathway check

IPEDS completions 2008–2019. Share of IPEDS graduates on LinkedIn, summed over school × field × year cells; crosswalk-collided schools dropped. Nonresident master's: 65% covered vs. 61% of other master's graduates. Foreign degrees (CtW) can't be benchmarked.

Key variables

- **Pathway**

- only-US-educated (USO): all degrees from US schools.
- Came-to-Study (CtS): first degree abroad, last degree in the US.
- Came-to-Work (CtW): all degrees abroad, then US work.

- **First job** First job starting after the degree; internships, volunteer work, TA/RA jobs, postdocs and “PhD student” titles don’t count.

- **School quality:** US admissions data (IPEDS) + QS rankings for foreign schools, on one scale.

- **Employer utility:** revealed-preference rankings from worker moves (Sorkin 2018).

Measuring mismatch

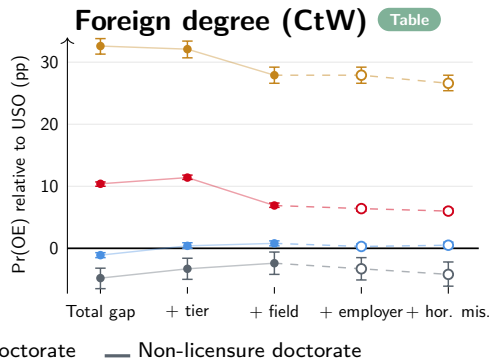
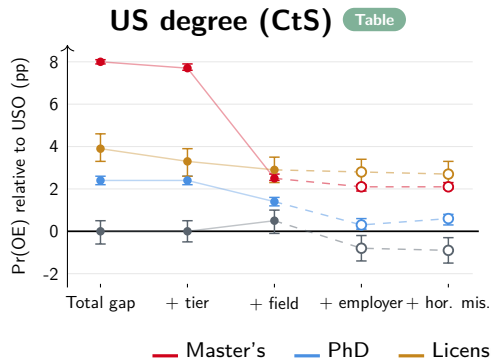
- **Over-educated:** your degree is above the typical entry requirement for your occupation (O*NET).
 -
 - USO over-education at first job: 82.9% (master's), 93.6% (PhD), 16.7% (licensure doc.), 99.1% (non-licensure doc.). 79% of master's OE matches are jobs that require at least a bachelor's degree.
 - So the object is the CtS/CtW - USO **gap** within degree, not the level.
- **Horizontally mismatched:** your job is not among the 20 most common jobs for your degree and field. Distribution
 - 41.4% of USO master's holders.
- **Occupation outcomes:** occupation median wage (OEWS) and revealed-preference rank, not individual pay.

Empirical Roadmap

- 1 Mismatch at Labor Market Entry
 - Horizontal mismatch
 - Occupational wage consequences
- 2 The Dynamics of the Friction
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 - Extensive margin: US placement at graduation
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Over-education patterns differ by terminal degree

First point: total gap. Each drop: share explained by the block added.



Linear probability models. Bars are 95% delta-method intervals using the full person-clustered coefficient covariance matrix. Field fixed effects are degree $\times$ field. Each line is the pathway main effect plus its degree interaction; total gap includes cohort fixed effects. Dashed, hollow steps condition on placement outcomes (employer, mismatch) and are descriptive.

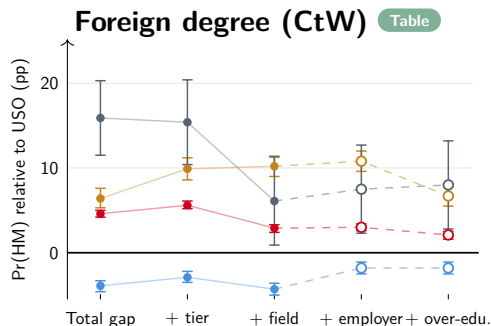
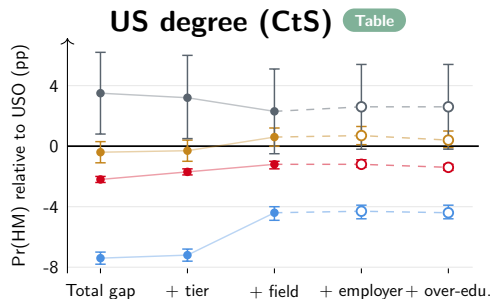
Master's holders and foreign licensure doctorates under-place; PhDs and non-licensure doctorates near parity

Table 1: Over-education gap vs. USO, same degree (pp)

Degree	Predicted		Observed	
	CtS	CtW	CtS	CtW
Master's	+	+	+2.5	+6.9
PhD	≈ 0	≈ 0	+1.4	+0.8
Non-licensure doctorate	≈ 0	≈ 0	+0.5	-2.4
Licensure doctorate	≈ 0	+	+2.9	+27.9

PhD holders are less likely to be horizontally mismatched

First point: total gap. Each drop: share explained by the block added.



— Master's — PhD — Licensure doctorate — Non-licensure doctorate

Linear probability models. Bars are 95% delta-method intervals using the full person-clustered coefficient covariance matrix. Field fixed effects are degree $\times$ field. Each line is the pathway main effect plus its degree interaction; total gap includes cohort fixed effects. Dashed, hollow steps condition on placement outcomes (employer, mismatch) and are descriptive.

Visa rules tie US-educated immigrants to their field; foreign-educated workers drift out of it

Table 2: Summary of horizontal mismatch at entry

Degree	CtS	CtW
MA	Matched	Mismatched
PhD	Matched	Matched
Non-licensure doctorate	More mismatched	Much more mismatched
Licensure doctorate	Weakly mismatched	More mismatched

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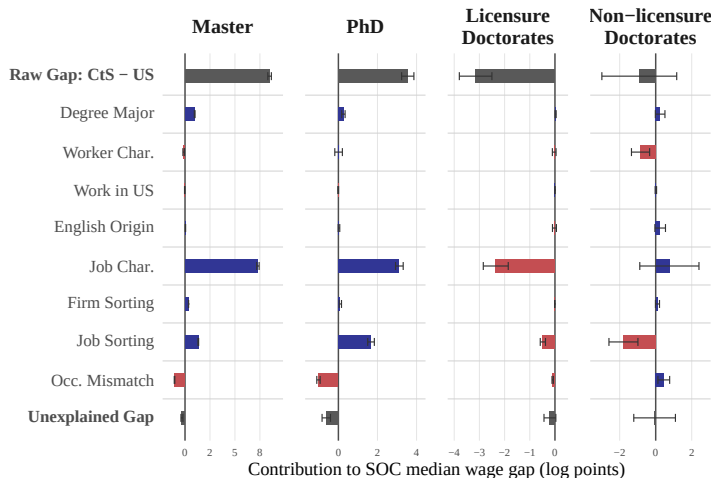
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CtS workers are less horizontally mismatched than CtW workers. OPT and H-1B both require a job related to the degree, so part of this gap is regulation, not friction.

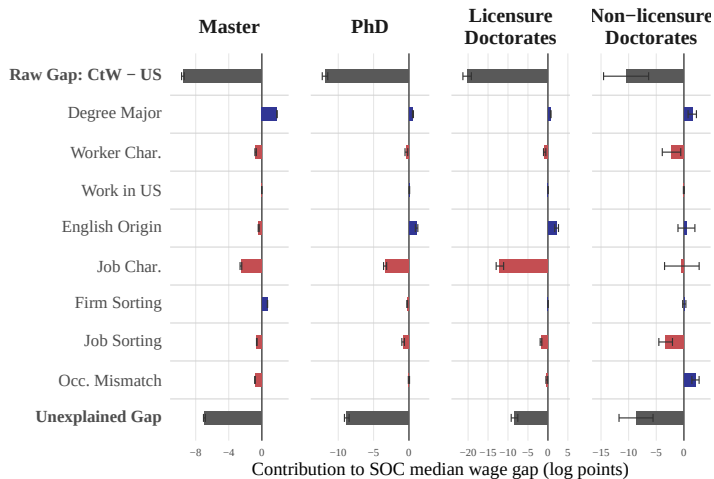
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US-educated master's sort into high-wage fields; mismatch claws back a small share



Foreign-trained workers retain large unexplained penalties



Gelbach decomposition of the first-US-job occupation median-wage gap. Job characteristics dominate explained gaps; unexplained penalties remain large.

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Measuring the gap in job quality: Oaxaca–Blinder

For each pathway p and degree e , split the gap in the occupation's median wage:

$$\bar{Y}_{pe} - \bar{Y}_{US,e} = \underbrace{(\bar{X}_{pe} - \bar{X}_{US,e})\hat{\beta}_{US,e}}_{\text{Explained}} + \underbrace{\bar{X}_{pe}(\hat{\beta}_{pe} - \hat{\beta}_{US,e})}_{\text{Unexplained}}$$

- X = things fixed **before** the job search: school tier, field, STEM, experience.
- **Explained:** immigrants have different school tiers, fields, etc.
- **Unexplained** $\approx -d +$ unobserved productivity difference. If immigrants are positively selected, a negative value is a lower bound on d ; a positive value says nothing about d .

Career dynamics distinguish attenuation from persistence

Design

- Separate Oaxaca decompositions at years 0 through 10 for stayers.

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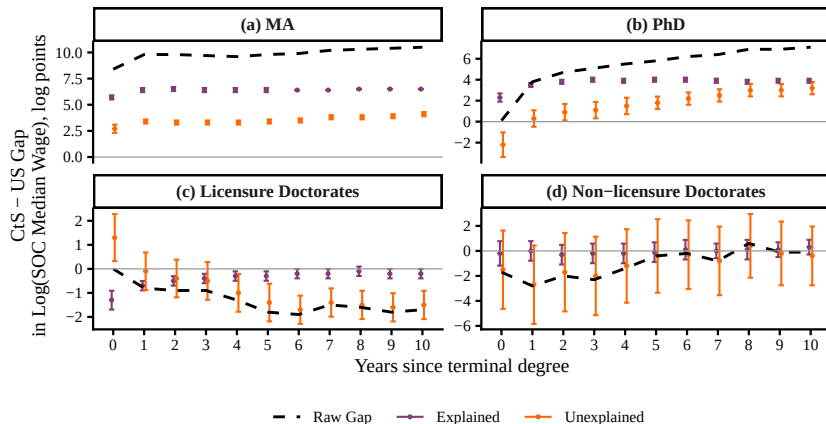
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- Changes in median wage (gaps) mostly from changing SOC's.
- Narrowing gaps are consistent with learning, assimilation, or expiring sponsorship obligations.
- Persistent gaps point toward credential, licensing, language, network, or taste-based discrimination barriers.

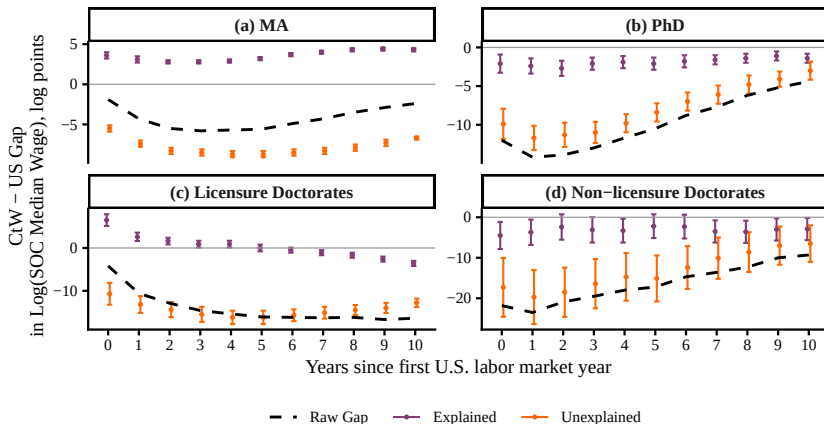
US credentials do not imply a common assimilation path



Notes: CtS includes only those who still work in the US at or after year 10. SOC median wage is the OEWS median wage of each year in 2015 dollars.

CtS workers observed in US jobs through year 10. The unexplained component is positive for master's holders (consistent with positive selection, uninformative about d), turns positive for PhDs, and remains negative for licensure doctorates.

Foreign-credential penalties remain large over the first decade



Notes: CtW includes only those who still work in the US at or after year 10. SOC median wage is the OEWS median wage of each year in 2015 dollars.

CtW workers observed in US jobs through year 10. Master's and regulated-doctorate penalties remain persistent; PhD penalties attenuate substantially.

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The 2008 STEM-OPT extension

Policy change

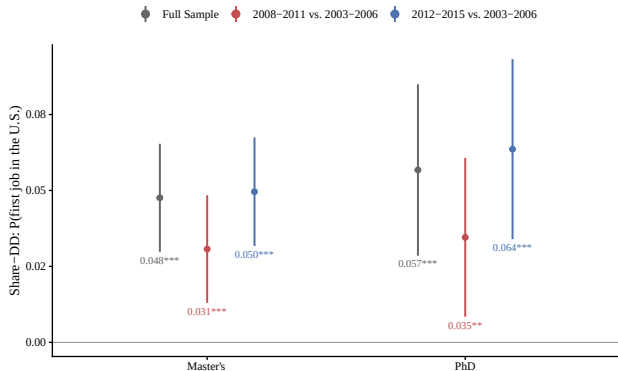
- April 2008 rule extended OPT from 12 to 29 months for F-1 STEM graduates.
- STEM graduates gained at least one additional H-1B filing season.
- Non-STEM graduates did not receive the extension.

Difference-in-differences Assumptions

- CtS master's and PhD graduates.
- STEM versus non-STEM majors.
- 2003–2006 comparison cohorts; 2007 excluded.
- Full, local, and post-recession windows.
- School tier and major generality controls.

The design estimates the effect of a specific reduction in hiring friction, not the level of a general friction.

STEM-OPT raised US entry but not retention



Came to Study. Extensive-margin share-DD on P(first job in the U.S.): [STEM post-pre] - [Non-STEM post-pre], covariate-adjusted (school tier and major generality); a two-way STEM x post difference on this single probability, not a triple difference. POSITIVE = STEM grads became relatively MORE likely to land a first job in the U.S. post-extension. This is exactly one minus the abroad-first outcome, estimated on the same sample. Labels = point estimate (*p<.05 **p<.01 ***p<.001). Pointranges 95% CI.

- **Entry:** master's +3.1 pp for 2008–11 graduates (enrolled before the rule), +5.0 for 2012–15 (after the recession). About 3 pp from the 2008 rule. 2012–15 cohorts' OPT windows overlap the 2016 extension to 36 months. No pre-trend. Event study
- **Retention:** share still in the US +0.2 pp at year 5, +0.6 at year 10 (neither significant).

CtS graduates; STEM = major on the 2008 designated-degree list; 2003–06 cohorts as the pre-period. Full sample treatment group is 2008–2021 graduation cohorts.

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Intensive margin: stayers' jobs five years after graduation

Stayers' jobs at year 5, STEM vs. non-STEM, post- vs. pre-2008 (master's):

	DD	95% CI
Over-educated (pp)	+1.4	[−1.0, 3.8]
Horizontally mismatched (pp)	−8.7	[−14.2, −3.2]
SOC Utility (log)	+0.17	[0.05, 0.30]
Firm Utility (log)	+0.30	[0.10, 0.50]
Occupation median wage (log)	+0.03	[−0.01, 0.06]

The gain is a **better employer within the same occupation**, a margin occupation-level over-education cannot see.

Increased job-lock

CtS master's still working in the US at year 5; 2008–21 vs. 2003–06 cohorts; CIs clustered by major. Extra staying is at most 1.3 pp (upper 95% bound), too small to produce the −8.7 pp by composition. PhDs: same pattern. PhD results

Changing only the work permit still moved jobs

- If d is a productivity gap, under-placement is efficient. If d is a pure cost, it is misallocation.
- The 2008 rule changed only H-1B lottery draws; 2008–11 graduates had already enrolled.
- Yet US entry rose 3 pp, and stayers reached more valuable employers.

So at least the part of d the rule removed was a cost, not a productivity gap.

Identifies only the part of d the policy removed. Horizontal-mismatch effects are not used here: visa rules require field-related jobs.

Employers' information or hiring cost?

Channel	Predicts	What I find	Evidence
Information <i>fades over time</i>			
Hiring cost <i>persists:</i> <i>sponsorship,</i> <i>licensing</i>			

[†] Fading is also consistent with sponsorship ending at green-card receipt.

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Hiring cost <i>persists: sponsorship, licensing</i>	A longer work permit raises US entry	Yes: +3 pp (about 1 in 7 would-be leavers)	causal (DD)

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Hiring cost <i>persists: sponsorship, licensing</i>	A longer work permit raises US entry ...and improves stayers' jobs	Yes: +3 pp (about 1 in 7 would-be leavers) Partly: MA hor. mis.↓, SOC utility ↑, firm utility ↑ at $t = 5$; over-education unchanged. Not from fewer leaving; stayers include new entrants	causal (DD) causal (DD)

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Channel	Predicts	What I find	Evidence
Information <i>fades over time</i>	Gap larger for foreign edu. Gap fades with US experience	Yes: CtW $\gg$ CtS at entry PhDs: yes. Master's: no	descriptive descriptive [†]
Hiring cost <i>persists: sponsorship, licensing</i>	A longer work permit raises US entry ...and improves stayers' jobs	Yes: +3 pp (about 1 in 7 would-be leavers) Partly: MA hor. mis.↓, SOC utility ↑, firm utility ↑ at $t = 5$; over-education unchanged. Not from fewer leaving; stayers include new entrants	causal (DD) causal (DD)

Both are present: information fades for PhDs; a hiring cost persists for master's, and cheaper hiring moves entry and placement. [†] Fading is also consistent with sponsorship ending at green-card receipt.

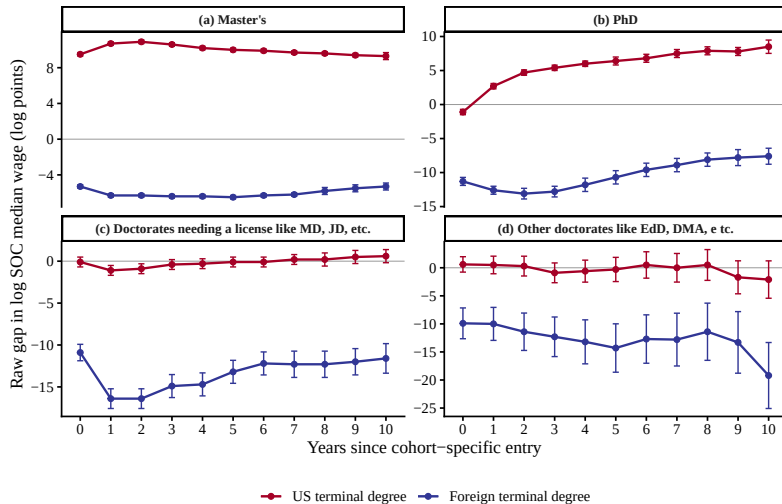
A friction firms can't price moves immigrants' jobs, not their pay

- ① A hiring friction that firms can't pass on through pay shows up as **job placement**: master's holders under-placed, PhDs near parity, foreign licensure doctorates blocked.
- ② The gap does **not** fade on a common path: foreign-degree master's holders stay behind for a decade.
- ③ Lowering the friction (STEM-OPT) raised US entry by about 3 pp and moved stayers to higher-ranked employers within their occupations; over-education did not change.

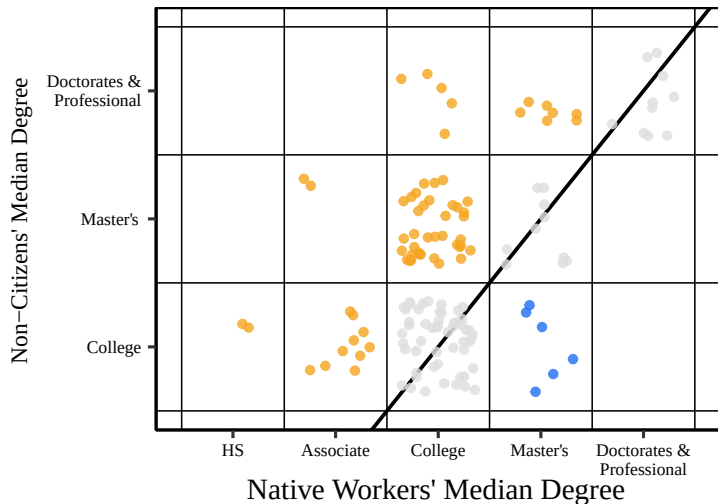
Questions?

Immigrants Career Paths in the US

Cross-sectional differences in median wage by pathway and degree



Median instead of modal degree

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Notation for the framework

$\theta, \bar{\theta}$	a worker's ability; $\bar{\theta}$ is the most able type
e	education attainment
$w(e)$	pay of the job typically held by someone with education e
$\bar{e}$	education the most able native chooses (think: master's)

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d	hiring friction: how much less an immigrant is worth to the firm
$\tilde{e}$	education typical of the job the immigrant actually gets ($< \bar{e}$)
$\hat{e}$	extra education an immigrant gets to reach a matching job (think: PhD)

Does LinkedIn coverage differ by pathway?

[◀ Back](#)

IPEDS reports US nonresident graduates, so I can check master's (2008–2019, 1,167 schools):

- LinkedIn covers **65%** of nonresident graduates and **61%** of other graduates.
- Within school $\times$ year, CtS profiles fall **25%** short of what classmates' coverage predicts.
- This is expected by construction: a nonresident whose degrees are all from the US is coded US-only.
 - The gap closes if **7–13%** of nonresident master's graduates hold a US bachelor's.
 - IPEDS flows: that requires **12–23%** of nonresident bachelor's graduates to stay for a US master's (vs. 15% of LinkedIn's mostly domestic bachelor's graduates).

IPEDS completions 2008-2019; LinkedIn master's matched to IPEDS unitid and CIP2; school-years with ≥ 50 IPEDS and ≥ 20 LinkedIn graduates; collided schools dropped. Range: excluding vs. keeping cells where LinkedIn US exceeds IPEDS residents. Ignores green-card holders and citizens with a foreign first degree, who work in the opposite direction.

Occupation coding

- Fine-tune Llama-3.1 (8B) on LinkUp's detailed O*NET job-title taxonomy.
- Classification requests one eight-digit SOC code and assigns invalid text to 00-0000.00.
- Match 74.3 million of 76.1 million unique titles to valid detailed SOC codes.
- Collapse to six-digit SOC codes for alignment with ACS and OEWS.
- Mean confidence score is 0.83.

Why it matters

Multilingual title coding retains career histories after workers move abroad, making exit observable when profiles are updated.

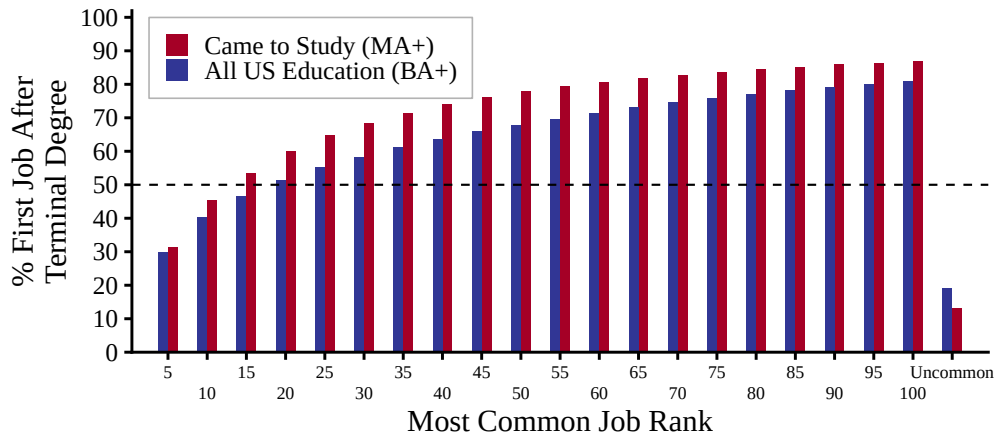
Institutional selectivity

- US institutions receive an admissions-based score using standardized test scores and rejection rates.
- Nine selectivity tiers range from the bottom quartile to the top 1%; unmatched institutions form tier 0.
- Foreign institutions receive the tier of the US institution with the closest average QS ranking.
- Regressions use flexible degree-by-tier indicators rather than a cardinal prestige assumption.

O*NET task components

- Principal component analysis uses the level scores of all 87 O*NET skills and abilities without ex ante categories.
- The first three components explain approximately 73% of total variation:
 - ① Cognitive and Interpersonal Competence
 - ② Attention and Coordination
 - ③ Practical Job Capabilities
- Certification importance and typical on-the-job training enter separately.
- Occupations with prevalent immigrant under-education are excluded because their mechanism differs from the model's over-education margin.

Distribution of first jobs among common jobs

[Back](#)

Sample coverage

Pipeline	Initial first jobs	Analytic sample	Retained
CtS–US graduate	8,873,285	4,758,933	53.6%
CtW–US	29,417,691	10,866,257	36.9%

- Common support requires institutional tier and occupation utility within degree-major.
- The CtW pipeline also requires arrival cohort, origin region, and non-US experience.
- The largest losses occur when credentials or occupations cannot be linked to external crosswalks.

Estimates apply to the covered population of credentialed workers and are not population-representative.

	(1)	(2)	(3)	(4)	(5)
	Pr(Over-Education)				
CtS	0.080*** (0.001)	0.077*** (0.001)	0.028*** (0.001)	0.024*** (0.001)	0.024*** (0.001)
CtS × PhD	-0.056*** (0.001)	-0.054*** (0.001)	-0.034*** (0.001)	-0.041*** (0.001)	-0.039*** (0.001)
CtS × Licensure Doc.	-0.041*** (0.003)	-0.044*** (0.003)	-0.000 (0.003)	0.004 (0.003)	0.003 (0.003)
CtS × Non-licensure Doc.	-0.080*** (0.003)	-0.077*** (0.003)	-0.029*** (0.003)	-0.037*** (0.003)	-0.040*** (0.003)
Hor. Mis.					0.054*** (0.000)
Baseline Pr. (USO, MA)	0.829	0.829	0.829	0.829	0.829
N	2,753,658	2,753,658	2,753,658	2,753,658	2,753,658

	(1)	(2)	(3)	(4)	(5)
	Pr(Over-Education)				
CtW	0.104*** (0.002)	0.114*** (0.002)	0.084*** (0.002)	0.078*** (0.002)	0.074*** (0.002)
CtW × Bachelor	-0.134*** (0.001)	-0.137*** (0.001)	-0.094*** (0.001)	-0.093*** (0.001)	-0.090*** (0.001)
CtW × PhD	-0.115*** (0.001)	-0.110*** (0.002)	-0.069*** (0.002)	-0.068*** (0.002)	-0.063*** (0.002)
CtW × Licensure Doc.	0.222*** (0.006)	0.206*** (0.007)	0.197*** (0.006)	0.203*** (0.006)	0.194*** (0.006)
CtW × Non-licensure Doc.	-0.152*** (0.008)	-0.148*** (0.009)	-0.090*** (0.010)	-0.093*** (0.010)	-0.112*** (0.011)
Hor. Mis.					0.117*** (0.000)
Baseline Pr. (USO, MA)	0.829	0.829	0.829	0.829	0.829
N	7,629,892	7,629,892	7,629,892	7,629,892	7,629,892

Immigrant PhDs reach parity at cap-exempt employers [◀ Back](#)

Over-education gap vs. same-degree USO workers (pp), by first employer

		Cap-exempt	Cap-subject	Difference
PhD	CtS	0.3 [−0.2, 0.9]	1.5 [1.3, 1.7]	−1.2 [−1.7, −0.6]
	CtW	−3.1 [−5.0, −1.3]	0.8 [0.4, 1.3]	−4.0 [−5.9, −2.0]
Master's	CtS	5.0 [4.1, 5.9]	2.3 [2.2, 2.5]	+2.7 [1.7, 3.6]
	CtW	5.0 [2.8, 7.1]	6.7 [6.4, 7.1]	−1.8 [−4.0, 0.4]

Without the H-1B lottery, the PhD contract is available: the PhD gap sits at cap-subject employers. Only 3% of CtS master's start at cap-exempt employers, plausibly campus jobs taken when cap-subject employers will not hire.

Column-3 specification with degree $\times$ field FE, estimated separately by the first employer's cap-exempt status; 95% CIs, person-clustered. Cap-exempt PhDs: 8,149 CtS, 9,521 CtW. The first employer is chosen in the same search, so the split locates the gap; it is not an effect of cap exemption.

CtS horizontal mismatch [◀ Back](#)

	(1)	(2)	(3)	(4)	(5)
	Pr(Horizontal mismatch)				
CtS	-0.022*** (0.001)	-0.017*** (0.001)	-0.012*** (0.001)	-0.011*** (0.001)	-0.013*** (0.001)
CtS × PhD	-0.052*** (0.002)	-0.055*** (0.002)	-0.036*** (0.002)	-0.037*** (0.003)	-0.033*** (0.003)
CtS × Licensure Doc.	0.018*** (0.004)	0.013*** (0.004)	0.018*** (0.003)	0.019*** (0.003)	0.018*** (0.003)
CtS × Non-licensure Doc.	0.057*** (0.014)	0.049*** (0.014)	0.047** (0.015)	0.048*** (0.015)	0.052*** (0.015)
Over Edu.					0.102*** (0.001)
Baseline Pr. (US, Master's)	0.414	0.414	0.414	0.414	0.414
N	2,753,658	2,753,658	2,753,658	2,753,658	2,753,658

CtW horizontal mismatch [◀ Back](#)

	(1)	(2)	(3)	(4)	(5)
	Pr(Horizontal mismatch)				
CtW	0.046*** (0.002)	0.056*** (0.002)	0.030*** (0.002)	0.031*** (0.002)	0.020*** (0.002)
CtW × Bachelor	-0.053*** (0.001)	-0.063*** (0.001)	-0.021*** (0.001)	-0.023*** (0.001)	-0.009*** (0.001)
CtW × PhD	-0.085*** (0.003)	-0.085*** (0.003)	-0.069*** (0.003)	-0.045*** (0.003)	-0.035*** (0.003)
CtW × Licensure Doc.	0.019** (0.006)	0.042*** (0.006)	0.074*** (0.006)	0.079*** (0.006)	0.049*** (0.006)
CtW × Non-licensure Doc.	0.113*** (0.022)	0.098*** (0.025)	0.148*** (0.026)	0.160*** (0.026)	0.174*** (0.026)
Over Edu.					0.147*** (0.000)
Baseline Pr. (US, Master's)	0.414	0.414	0.414	0.414	0.414
N	7,629,892	7,629,892	7,629,892	7,629,892	7,629,892

Identifying the friction at entry

For pathway p and terminal-degree group e , decompose the occupation median-wage gap:

$$\bar{Y}_{pe} - \bar{Y}_{US,e} = \underbrace{(\bar{X}_{pe} - \bar{X}_{US,e})\hat{\beta}_{US,e}}_{\text{explained sorting}} + \underbrace{\bar{X}_{pe}(\hat{\beta}_{pe} - \hat{\beta}_{US,e})}_{\text{unexplained } \approx -d_e + \text{selection}}.$$

- Y is the residualized log OEWS median wage of the occupation held.
- X includes pre-market education, origin, and experience characteristics in the preferred friction estimate.
- Gelbach decompositions then attribute the raw gap to worker, job, firm, sorting, and mismatch blocks.

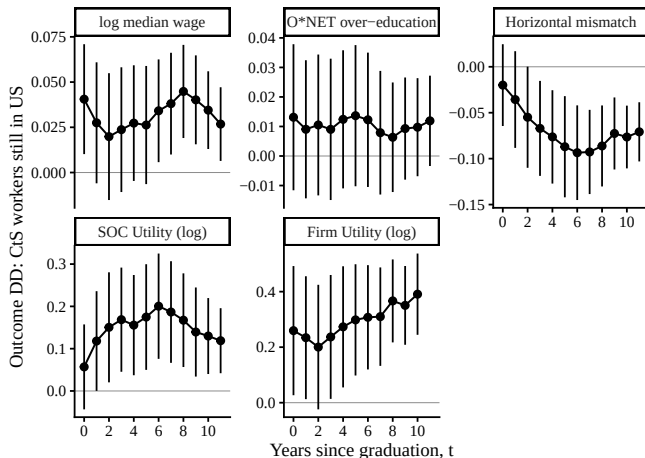
Lower-bound interpretation

Current occupation is downstream of the friction. Conditioning on placement-related covariates can absorb part of the object the model seeks to measure.

- ① **Parallel trends:** absent the rule, STEM and non-STEM foreign-placement rates would have evolved similarly.
- ② **No differential concurrent shock:** the recession did not independently improve STEM prospects relative to non-STEM prospects.
- ③ **Stable treatment composition:** students did not immediately re-sort into STEM in response to the policy.
- ④ **Monotone selection:** the extension did not cause a graduate who otherwise would have stayed to leave.

The 2008–2011 window limits program-choice responses but overlaps the recession.
The 2012–2015 window avoids the recession but permits endogenous major choice.
The two windows address different threats.

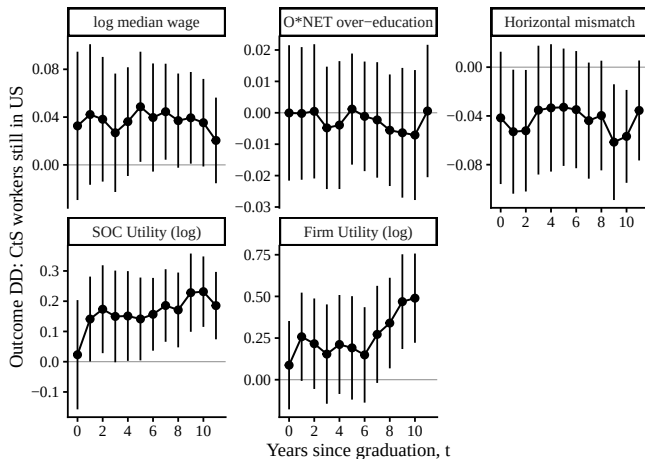
STEM-OPT job-quality estimates: master's



Came to Study, Master's. Common-horizon DD of the CtS workers still in US OUTCOME gap: [STEM post-pre] - [Non-STEM post-pre]. Points range 95% CI.

Difference-in-differences estimates among CtS master's graduates still working in the United States.

STEM-OPT job-quality estimates: PhD



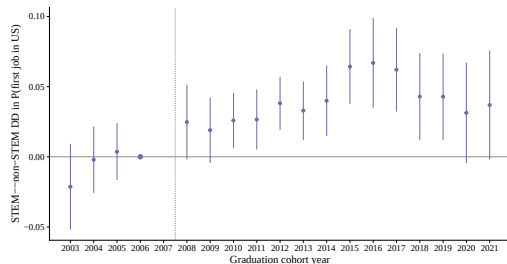
Came to Study, PhD. Common-horizon DD of the CtS workers still in US OUTCOME gap: [STEM post-pre] - [Non-STEM post-pre]. Point ranges 95% CI.

Difference-in-differences estimates among CtS PhD graduates still working in the United States.

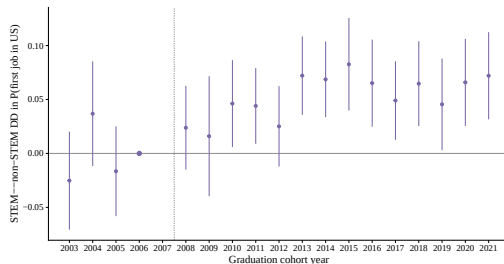
The policy's effect on foreign-degree workers [◀ Back](#)

- CtW workers never had student visas, so OPT doesn't cover them. But they are **not a placebo**:
 - they draw from the same H-1B lottery; a 29-month OPT gives STEM CtS graduates up to 3 draws instead of 1;
 - the 20,000 advanced-degree slots are for US degrees only;
 - if employers see CtS and CtW as substitutes, cheaper CtS hiring lowers demand for CtW.
- Prediction: STEM CtW placement and job quality **fall** relative to non-STEM after 2008.
- Outcomes: STEM share of CtW arrivals by arrival year; job quality at arrival; a triple difference against CtS.

Cohort event study, US placement

[◀ Back](#)

Notes: Came to Study, Master's. Outcome: first resolved job is in the US. STEM = own terminal-degree major on the 2008 DHS STEM list. Each point is the STEM—non-STEM difference-in-differences for a graduation cohort relative to 2006 (normalized to zero). Cohorts 2003—2005 provide the pre-trend diagnostic; 2007 is omitted because STEM—OPT eligibility is ambiguous at the policy boundary. Error bars are 95% CRV1 confidence intervals clustered by normalized major.



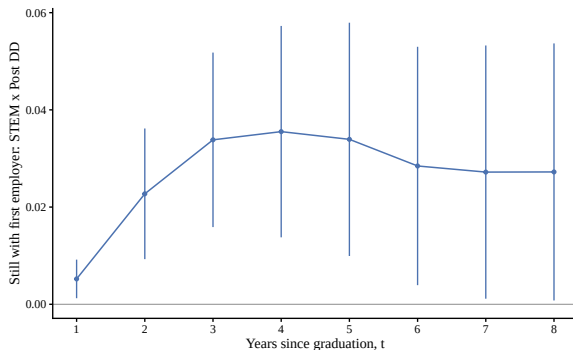
Notes: Came to Study, PhD. Outcome: first resolved job is in the US. STEM = own terminal-degree major on the 2008 DHS STEM list. Each point is the STEM—non-STEM difference-in-differences for a graduation cohort relative to 2006 (normalized to zero). Cohorts 2003—2005 provide the pre-trend diagnostic; 2007 is omitted because STEM—OPT eligibility is ambiguous at the policy boundary. Error bars are 95% CRV1 confidence intervals clustered by normalized major.

Master's (left): pre-period $-2.1, -0.2, +0.4$; 2008–11 $+1.9$ to $+2.7$; 2015–17 $+6.4$ to $+6.7$ (the 2016 rule).

PhD (right): significant from the 2010 cohort on.

Reference cohort 2006; 2007 omitted. 95% CIs clustered by major.

STEM master's stay longer with the first employer

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Still with the first employer: +3.4 to +3.6 pp at years 3–5. Two interpretations: fewer forced attrition after a failed first petition (predicts years 1–2), or waiting out the lottery (builds over years 1–3). The path is closer to the second. PhDs: no effect.

Cap-exempt first employers

- **Destination:** cap-exempt share of US first jobs unchanged: 0.0 pp (master's), -0.9 pp (PhD; 95% CI -4.7 to $+2.9$).
- **Master's:** cap-subject starters (85%) match the full sample. Cap-exempt: too few (CI on mismatch -11.3 to $+10.4$).
- **PhDs:** gains are *not* smaller at cap-exempt employers: occupation wage $+0.09$ vs. $+0.03$, rank $+0.21$ vs. $+0.14$ (year 5).
- Likely selection: the split conditions on where graduates chose to start, which the policy may have changed.